

Smart Campus Operations Hub Feature Distribution for 4-Member Group

Member 1 – Facilities & Assets Management

Responsible for managing resources like rooms, labs, and equipment.

- 1 Resource Catalogue Management
- 2 Resource Metadata Management
- 3 Search & Filtering
- 4 Build CRUD APIs for resources
- 5 Develop UI for listing, adding, editing resources

Member 2 – Booking Management System

Responsible for booking workflows and conflict handling.

- 1 Booking Request System
- 2 Booking Workflow (Pending → Approved/Rejected → Cancelled)
- 3 Conflict Detection (avoid overlapping bookings)
- 4 Booking Viewing System
- 5 Develop booking APIs and admin approval UI

Member 3 – Maintenance & Incident System

Handles issue reporting and ticket lifecycle.

- 1 Ticket Creation System
- 2 Image Attachments (up to 3 images)
- 3 Ticket Workflow (Open → In Progress → Resolved → Closed)
- 4 Technician Assignment
- 5 Comment System
- 6 Develop ticket APIs and UI

Member 4 – Authentication, Notifications & Security

Responsible for system security and communication features.

- 1 OAuth Login (Google)

- 2 Role-Based Access Control (USER, ADMIN, etc.)
- 3 Notification System
- 4 Secure API endpoints
- 5 Protected frontend routes
- 6 Notification UI panel

Final Notes

- 1 Each member should implement at least 4 REST endpoints
- 2 Ensure both frontend and backend contributions
- 3 Maintain proper Git commits per member
- 4 Be ready to explain your module during viva